Network Protocols Assignment

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Task

Fill the table below with information about top 100 network protocols.

Table

#	Protocol	Layer	Port	TCP/UDP	Encryption	Usage
1	HTTP	Application	80	TCP	No	Web browsing
2	HTTPS	Application	443	TCP	Yes	Secure web
3	FTP	Application	20,21	TCP	No	File transfer
4	SFTP	Application	22	TCP	Yes	Secure file transfer
5	FTPS	Application	989,990	TCP	Yes	FTP over SSL
6	TFTP	Application	69	UDP	No	Simple file transfer
7	SSH	Application	22	TCP	Yes	Remote login
8	Telnet	Application	23	TCP	No	Remote login
9	SMTP	Application	25, 587, 2525	TCP	No	Email sending
10	POP3	Application	110,995	TCP	No	Email retrieval
11	IMAP	Application	143,993	TCP	No	Email access
12	DNS	Application	53	UDP/TCP	No	Name resolution

13	DHCP	Application	67,68	UDP	No	IP assignment
14	SNMP	Application	161,162	UDP	Yes (v3)	Monitoring
15	RDP	Application	3389	TCP	Yes	Remote desktop
16	VNC	Application	5900	TCP	No	Remote desktop
17	NTP	Application	123	UDP	No	Time sync
18	LDAP	Application	389	TCP/UDP	No	Directory
19	LDAPS	Application	636	TCP	Yes	Secure directory
20	SMB	Application	445	TCP	No	File sharing
21	NFS	Application	2049	TCP/UDP	No	File sharing
22	AFP	Application	548	TCP	No	Apple sharing
23	BGP	Network	179	TCP	No	Routing
24	RADIUS	Application	1812,1813	UDP	No	AAA
25	TACACS+	Application	49	TCP	Yes	AAA
26	Kerberos	Application	88	TCP/UDP	Yes	Authentication
27	SIP	Application	5060,5061	TCP/UDP	No	VoIP
28	RTP	Application	Dynamic	UDP	No	Media
29	RTCP	Application	Dynamic	UDP	No	Media control
30	RTSP	Application	554	TCP/UDP	No	Streaming
31	H.323	Application	1720	TCP	No	VoIP
32	Syslog	Application	514	UDP	No	Logging

33	Whois	Application	43	TCP	No	Domain lookup
34	IRC	Application	6667	TCP	No	Chat
35	XMPP	Application	5222	TCP	No	Messaging
36	NNTP	Application	119	TCP	No	News
37	MQTT	Application	1883,8883	TCP	Yes	IoT
38	CoAP	Application	5683	UDP	No	IoT
39	WebRTC	Application	Dynamic	UDP	Yes	Real-time comm
40	STUN	Application	3478	UDP	No	NAT traversal
41	TURN	Application	3478	UDP	No	Relay
42	BitTorrent	Application	6881-6889	TCP/UDP	No	P2P
43	Rsync	Application	873	TCP	No	File sync
44	WebSocket	Application	80,443	TCP	Yes	Full duplex
45	IPsec	Network	N/A	IP	Yes	VPN
46	IKE	Application	500	UDP	Yes	Key exchange
47	ESP	Network	N/A	IP	Yes	Encryption
48	AH	Network	N/A	IP	No	Auth only
49	PPTP	Data Link	1723	TCP	No	VPN
50	L2TP	Data Link	1701	UDP	No	VPN
51	OpenVPN	Application	1194	UDP/TCP	Yes	VPN
52	WireGuard	Network	51820	UDP	Yes	VPN
53	TLS	Application	N/A	TCP	Yes	Security
54	SSL	Application	N/A	TCP	Yes	Deprecated

55	EAP	Data Link	N/A	N/A	Yes	Auth
56	TCP	Transport	N/A	N/A	No	Reliable transport
57	UDP	Transport	N/A	N/A	No	Fast transport
58	SCTP	Transport	N/A	N/A	No	Multi-stream
59	DCCP	Transport	N/A	N/A	No	Congestion ctrl
60	QUIC	Transport	443	UDP	Yes	Fast web
61	ICMP	Network	N/A	IP	No	Ping
62	IGMP	Network	N/A	IP	No	Multicast
63	OSPF	Network	N/A	IP	No	Routing
64	EIGRP	Network	N/A	IP	No	Routing
65	RIP	Network	520	UDP	No	Routing
66	IS-IS	Network	N/A	N/A	No	Routing
67	ARP	Data Link	N/A	N/A	No	IP→MAC
68	RARP	Data Link	N/A	N/A	No	MAC→IP
69	NDP	Network	N/A	ICMPv6	No	IPv6
70	HSRP	Network	1985	UDP	No	Redundancy
71	VRRP	Network	N/A	IP	No	Redundancy
72	GLBP	Network	3222	UDP	No	Load balance
73	STP	Data Link	N/A	N/A	No	Loop prevention
74	RSTP	Data Link	N/A	N/A	No	Fast STP
75	LLDP	Data Link	N/A	N/A	No	Discovery
76	CDP	Data Link	N/A	N/A	No	Cisco discovery
77	VTP	Data Link	N/A	N/A	No	VLAN

78	LACP	Data Link	N/A	N/A	No	Link agg
79	Ethernet	Data Link	N/A	N/A	No	LAN
80	Wi-Fi	Data Link	N/A	N/A	Yes	WLAN
81	PPP	Data Link	N/A	N/A	No	Point link
82	PPPoE	Data Link	N/A	N/A	No	DSL
83	HDLC	Data Link	N/A	N/A	No	WAN
84	Frame Relay	Data Link	N/A	N/A	No	WAN
85	MPLS	Data Link	N/A	N/A	No	Forwarding
86	VXLAN	Data Link	4789	UDP	No	Virtual LAN
87	GRE	Network	N/A	IP	No	Tunneling
88	LISP	Network	4341	UDP	No	Routing
89	OpenFlow	Application	6653	TCP	Yes	SDN
90	NETCONF	Application	830	TCP	Yes	Config
91	RESTCONF	Application	443	TCP	Yes	Config
92	gRPC	Application	N/A	TCP	Yes	API
93	AMQP	Application	5672	TCP	No	Messaging
94	MySQL	Application	3306	TCP	No	Database
95	PostgreSQL	Application	5432	TCP	No	Database
96	MongoDB	Application	27017	TCP	No	NoSQL
97	Redis	Application	6379	TCP	No	Cache
98	Memcached	Application	11211	TCP/UDP	No	Cache
99	Gopher	Application	70	TCP	No	Old web
100	Finger	Application	79	TCP	No	User info